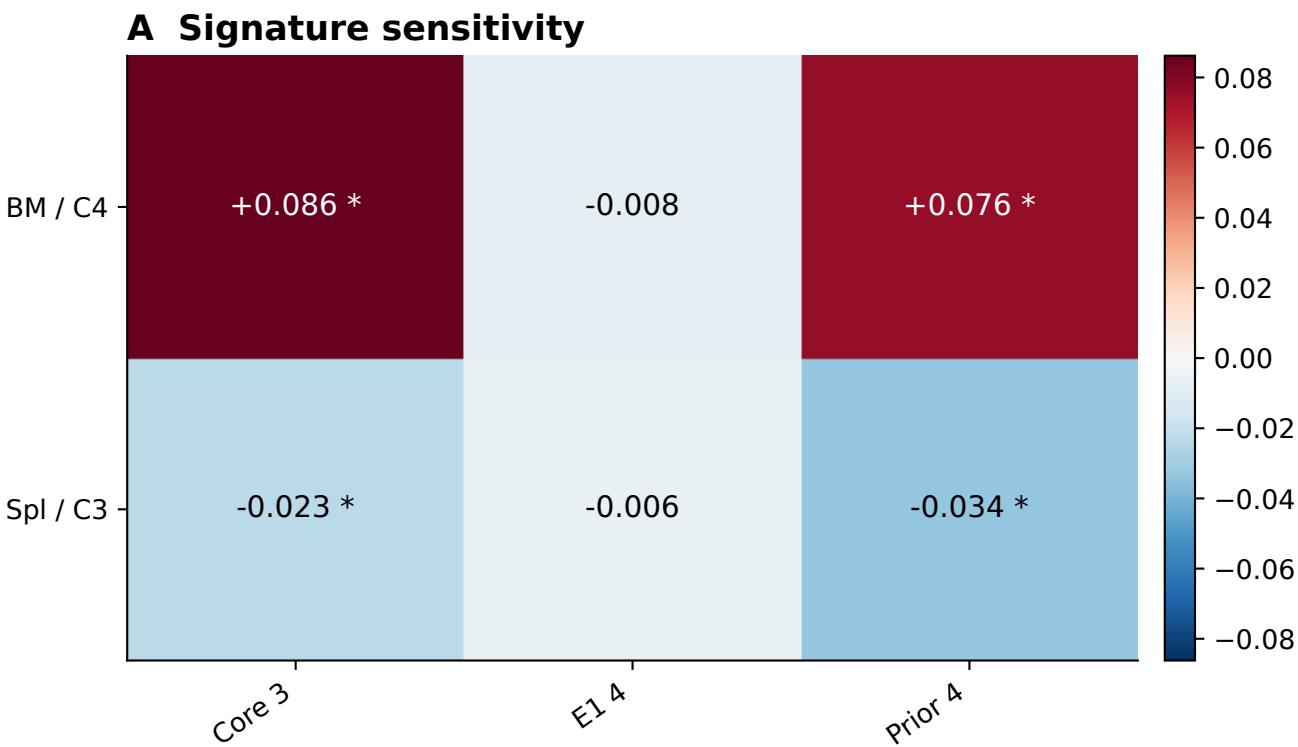


Core signature changes do not imply a full killing pathway

Two reproducible local core signals; the broader E1 signature is not significant.



B Complementary evidence from the same dataset

Bone marrow C4 core: +0.086; q = 0.025
Spleen C3 core: -0.023; q = 0.0015
Both: same direction in all 5 QC resamples

Stable-cluster effects: +0.111 and -0.022
Original-cell overlap: 98.7% and 99.5%

E1 (+Ctla2a): q = 0.93 and 0.53
Killing-related GO terms: no q ≤ 0.05

The evidence supports a localized core-effector transcriptional shift.
It does not establish increased/decreased killing activity or coordinated activation of a full cytotoxic pathway.